Module Code: CO661

Examination number: 199662

University of Kent

May/June 2021

Examination Answer Booklet

Before starting this examination, you **MUST enter your Examination Number** above **(double click)**; failure to provide this means your mark cannot be recorded and **you may be awarded a 0 (zero)** for this examination.

Re-save the answer booklet to include your exam number in the file name, using the following format: [**your exam number]** [**module code] AnswerBooklet** (e.g. *123456 XX123 AnswerBooklet*).

Answer one question per page. You may add additional pages to this document if necessary.

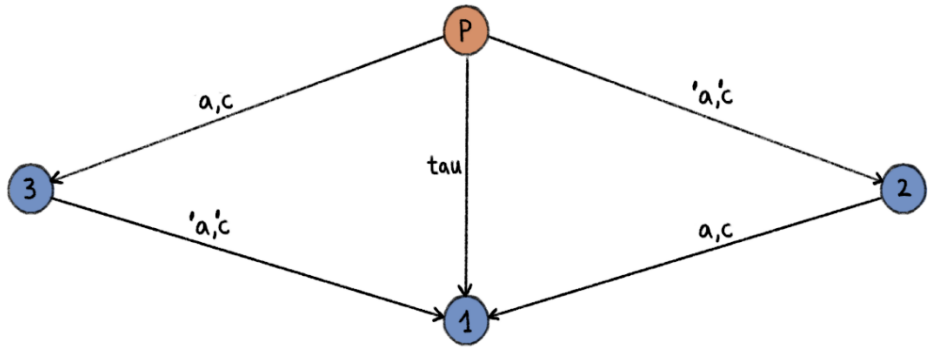
Question Number	1a

Add your answer below:

$$\text{handshake}\ \frac{\nu a \xrightarrow{a} 0 \qquad \nu b \xrightarrow{\bar{a}} 0}{\nu a, \nu b\,.\,((a.\,\mathbf{0} + b.\,\mathbf{0}) \mid \bar{a}.\,\mathbf{0}) \xrightarrow{\tau} 0}$$

Question Number	1b

Add your answer below:



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P = (′ a(bar).0 + ′ c(bar).0) | (a.0 + c.0)

1 = 0

2 = a.0 + c.0

3 = ′ a(bar).0 + ′ c(bar).0

Question Number	1c

Add your answer below:

i) Client = syn(bar).syn-ack.ack(bar).0

ii)

iii)

Question Number	2a

Add your answer below:

Question Number	2b

Add your answer below:

i) Yes, because of compensation.

ii)

iii)

Question Number	4a

Add your answer below:

i) As it has no finite witness, due to the dates being unknown, it is a liveliness property. To make it a safety property, it can be modified to say: “Your goods will be dispatched on December 2nd, 2021”.

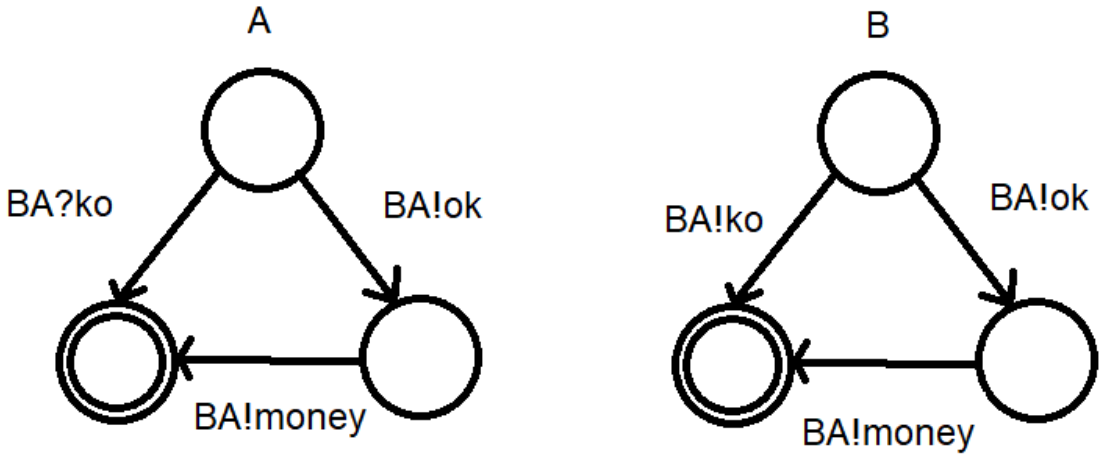
ii) As it has no finite witness, as there is no determination of when the lamp will change between the red and green pulses, it is a liveliness property. To make it a safety property, it can be modified to say: “The lamp will send an alternation every of “red” and “green” pulses every 5 seconds”.

Question Number	4b

Add your answer below:

i) {BA!ok, BA?ok, BA!money, BA?money}
{BA!ok, BA!money, BA?ok, BA?money}
{BA!ok, BA!money, BA?money, BA?ok}
{BA!ko, BA?ko}

ii) Fininte state machine R:



iii) There are no traces that is produced in R that cannot be found in M.

iv) {MSC1, MSC1} is realizable, because all traces of finite state machine R are found in finite state machine M.